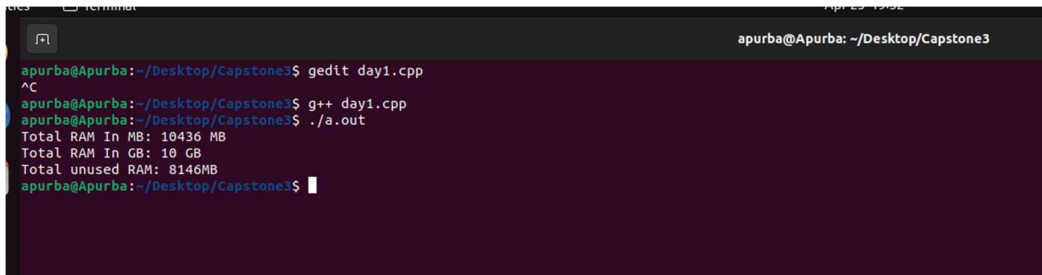


SYSTEM MONITOR TOOL

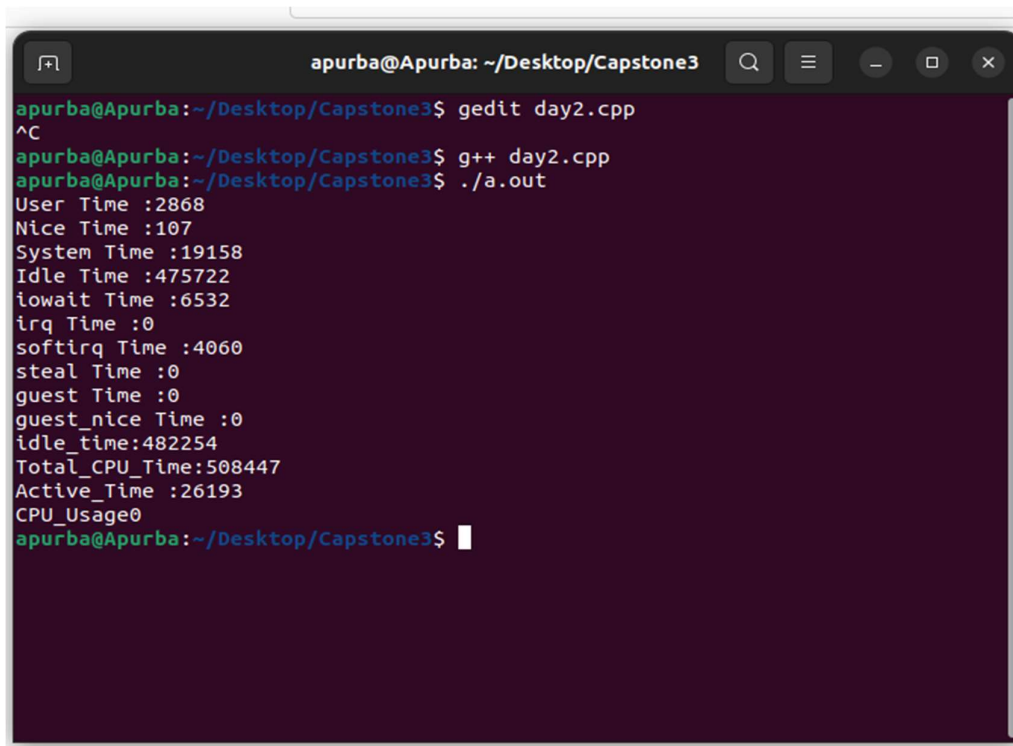
OBJECTIVE:- Create a system monitor tool in C++ that displays real time information about system processes, memory usage ,and CPU load ,similar to 'TOP' command.

DAY1: DESIGN UI LAYOUT AND GATHER SYSTEM DATA USING SYSTEM CALLS

A terminal window with a dark purple background. The prompt is 'apurba@Apurba: ~/Desktop/Capstone3'. The user enters 'gedit day1.cpp', followed by '^C' to cancel. Then they enter 'g++ day1.cpp' and './a.out'. The output shows memory statistics: 'Total RAM In MB: 10436 MB', 'Total RAM In GB: 10 GB', and 'Total unused RAM: 8146MB'.

```
apurba@Apurba:~/Desktop/Capstone3$ gedit day1.cpp
^C
apurba@Apurba:~/Desktop/Capstone3$ g++ day1.cpp
apurba@Apurba:~/Desktop/Capstone3$ ./a.out
Total RAM In MB: 10436 MB
Total RAM In GB: 10 GB
Total unused RAM: 8146MB
apurba@Apurba:~/Desktop/Capstone3$
```

DAY2: DISPLAY PROCESS LIST WITH CPU AND MEMORY USAGE .

A terminal window with a dark purple background, titled 'apurba@Apurba: ~/Desktop/Capstone3'. The user enters 'gedit day2.cpp', followed by '^C'. Then they enter 'g++ day2.cpp' and './a.out'. The output displays various system statistics including user time, nice time, system time, idle time, iowait time, irq time, softirq time, steal time, guest time, guest_nice time, idle_time, Total_CPU_Time, Active_Time, and CPU_Usage0.

```
apurba@Apurba:~/Desktop/Capstone3$ gedit day2.cpp
^C
apurba@Apurba:~/Desktop/Capstone3$ g++ day2.cpp
apurba@Apurba:~/Desktop/Capstone3$ ./a.out
User Time :2868
Nice Time :107
System Time :19158
Idle Time :475722
iowait Time :6532
irq Time :0
softirq Time :4060
steal Time :0
guest Time :0
guest_nice Time :0
idle_time:482254
Total_CPU_Time:508447
Active_Time :26193
CPU_Usage0
apurba@Apurba:~/Desktop/Capstone3$
```

DAY3: IMPLEMENT PROCESS SORTING BY CPU AND MEMORY USAGE

```
apurba@Apurba: ~/Desktop/Capstone3
apurba@Apurba:~/Desktop/Capstone3$ gedit day3.cpp
^C
apurba@Apurba:~/Desktop/Capstone3$ g++ day3.cpp
apurba@Apurba:~/Desktop/Capstone3$ ./a.out
PID      CPU%    Memory (kB)   Name
2794     22.5633 426436        (gnome-shell)
3784      2.5641  3712          (a.out)
3165      1.9439  26904         (pulseaudio)
707       1.67997 0              (dbus-daemon)
2620      1.57251 10356         (dbus-daemon)
726       1.48446 0              (snapd)
718       1.45549 0              (polkitd)
1         1.03093 0              (systemd)
90        0.83247 0              (kworker/u24:1-events_power_efficient)
708       0.814642 0              (NetworkManager)
apurba@Apurba:~/Desktop/Capstone3$
```

DAY4: ADD FUNCTIONALITY TO KILL PROCESS

```
apurba@Apurba: ~/Desktop/Capstone3
apurba@Apurba:~/Desktop/Capstone3$ gedit day4.cpp
apurba@Apurba:~/Desktop/Capstone3$ g++ day4.cpp
apurba@Apurba:~/Desktop/Capstone3$ ./a.out
PID      CPU%    Memory (kB)   Name
4998     72.1005 0              (unattended-upgr)
2794     25.0216 461380        (gnome-shell)
3882     15.5027 399288        (firefox)
5138     12.0714 10496         (http)
5137     10.3929 10240         (http)
5147      2        3712          (a.out)
83       1.57673 0              (ksoftirqd/11)
4003     0.988827 0              (kworker/u24:3-events_unbound)
3165     0.970037 27132         (pulseaudio)
5055     0.912698 68988         (gedit)
Enter PID to Kill :
3882
Process 3882 terminated successfully
apurba@Apurba:~/Desktop/Capstone3$
```

DAYS:IMPLEMENT REAL-TIME UPDATE FEATURE TO REFRESH DATA EVERY FEW SECONDS

```
apurba@Apurba: ~/Desktop/Capstone3
PID    CPU%    Memory (kB)    Name
3509    94.1713  343396         (firefox)
2820    59.556   381464         (gnome-shell)
4082    8.51668  101272         (WebExtensions)
3169    7.01524  26356          (pulseaudio)
2630    5.88752  11392          (dbus-daemon)
4444    5.55556  3584           (a.out)
714     4.69977  0              (dbus-daemon)
725     4.16539  0              (polkitd)
2622    3.29925  11136          (snapd-desktop-i)
441     2.57616  0              (kworker/u24:12-ext4-rsv-conversion)
Enter PID to Kill :
3509
Process 3509 terminated successfully
Press 'q' to quit or Enter to refresh :
```

After refreshing :-

```
apurba@Apurba: ~/Desktop/Capstone3
PID    CPU%    Memory (kB)    Name
2820    42.4129  390372         (gnome-shell)
3169    4.11582  26732          (pulseaudio)
2630    3.43876  11392          (dbus-daemon)
714     3.21988  0              (dbus-daemon)
725     2.82763  0              (polkitd)
2622    1.89868  11136          (snapd-desktop-i)
441     1.83799  0              (kworker/u24:12-events_unbound)
733     1.62771  0              (snapd)
1       1.61364  0              (systemd)
439     1.55133  0              (kworker/u24:10-flush-8:0)
Enter PID to Kill :

```

